

FORMULAS + KEY INFO

MATH

geometry

s $\left\{ \begin{array}{l} \text{area} = s^2 \\ \text{perimeter} = 4s \end{array} \right.$
 l, w $\left\{ \begin{array}{l} \text{area} = lw \\ \text{perimeter} = 2l + 2w \end{array} \right.$
 $\frac{b_1}{h}, \frac{b_2}{h}$ $\left\{ \begin{array}{l} \text{area} = \frac{b_1 + b_2}{2} h \end{array} \right.$
 $n = \# \text{ sides}$ $\left\{ \begin{array}{l} \sum \text{angles} = 180(n-2) \\ \text{single angle} = \frac{180(n-2)}{n} \end{array} \right.$

r $\left\{ \begin{array}{l} \text{area} = \pi r^2 \\ \text{circumference} = 2\pi r \\ \text{central angle} \rightarrow \angle x = 2\angle z \\ \text{arc length} = \frac{x}{360} 2\pi r \\ \text{sector area} = \frac{x}{360} \pi r^2 \end{array} \right.$
 c $\left\{ \begin{array}{l} \text{diameter} \rightarrow c \\ \text{chord} \rightarrow b, h \\ \text{pythagorean} \rightarrow h^2 + b^2 = c^2 \\ \text{area} = \frac{hb}{2} \end{array} \right.$
 supplementary $\rightarrow$ angles sum to 180°
 complementary $\rightarrow$ angles sum to 90°
 parallel lines w/ intersection:
 congruent angles
 same color
 $\angle x + \angle z = 180^\circ$
 $\angle z + \angle z = 180^\circ$

r, h $\left\{ \begin{array}{l} \text{surface area} = 2\pi r^2 + 2\pi rh \\ \text{volume} = \pi r^2 h \end{array} \right.$
 special right triangles: $\left\{ \begin{array}{l} 30^\circ, 60^\circ, 90^\circ \rightarrow 1: \sqrt{3}: 2 \\ 45^\circ, 45^\circ, 90^\circ \rightarrow 1: 1: \sqrt{2} \end{array} \right.$ **RATIO!**
 perpendicular lines $\rightarrow$ neg. reciprocal slopes
 || parallel lines $\rightarrow$ same slope

s $\left\{ \begin{array}{l} \text{surface area} = 6s^2 \\ \text{volume} = s^3 \\ \text{diagonal} = s\sqrt{3} \end{array} \right.$
 triangle exterior angle: $\left\{ \begin{array}{l} x + y = z \end{array} \right.$
 isosceles triangle: $\left\{ \begin{array}{l} x = z \end{array} \right.$
 equilateral triangle: $\left\{ \begin{array}{l} x = z = 60^\circ \end{array} \right.$

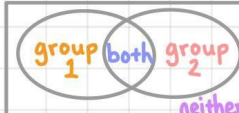
l, w, h $\left\{ \begin{array}{l} \text{surface area} = 2lw + 2lh + 2wh \\ \text{volume} = lwh \\ \text{diagonal} = \sqrt{l^2 + w^2 + h^2} \end{array} \right.$
 distance b/w $= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
 midpoint $= \left(\frac{x_2 + x_1}{2}, \frac{y_2 + y_1}{2} \right)$
 quadrants: $\left(\begin{array}{cc} (-, +) & (+, +) \\ (-, -) & (+, -) \end{array} \right)$

probability

combinations: $nCr = \frac{n!}{r!(n-r)!}$
 order not matter
 permutations: $nPr = \frac{n!}{(f_1!)(f_2!)(f_3!) \dots}$
 order matter
 when items chosen indistinguishable
 $n \rightarrow$ total, $r \rightarrow$ chosen, $f \rightarrow$ frequency of each event
 counting principle: total possible outcomes is product of all ways of each event occurring
 examples:

a) how many ways arrange letters ABBCCC
 $n = 6$
 $A \rightarrow f_1 = 1$
 $B \rightarrow f_2 = 2$
 $C \rightarrow f_3 = 3$
 $nPr = \frac{6!}{(1!)(2!)(3!)} = 120$ indistinguishable
 c) ways chose 3 people from group of 5
 $nCr = \frac{5!}{3!2!} = \frac{5 \times 4 \times 3 \times 2 \times 1}{3 \times 2 \times 1 \times 2 \times 1} = 10$
 $3 \times 2 \times 1$ cancel out

b) how many ways arrange 5 english & 3 science books on shelf so that subjects are together
 2 placement options $nPr = \frac{2!}{0!} = 2!$
 $\left\{ \begin{array}{l} S \rightarrow nPr = \frac{3!}{0!} = 3! \\ E \rightarrow nPr = \frac{5!}{0!} = 5! \end{array} \right.$ (2!)(3!)(5!) counting principle

complementary events: $P(A) + P(\text{not } A) = 1$
 independent: $P(A \& B) = P(A)P(B)$
 dependent: $P(A \& B) = P(A)P(A|B)$
 $\hookrightarrow P(A|B) \rightarrow$ probability of B given event A
 mutually exclusive: $P(A \text{ or } B) = P(A) + P(B)$
 mutually inclusive: $P(A \text{ or } B) = P(A) + P(B) - P(A \& B)$
 $\left\{ \begin{array}{l} P(A \cup B) \end{array} \right.$

 total = group 1 + group 2 - both + neither
 only group 1 = group 1 - both

examples:
 a) draw queen or heart
 $P(Q \text{ or } H) = P(Q) + P(H) - P(Q \& H)$
 b) draw queen or king
 $P(Q \text{ or } K) = P(Q) + P(K)$
 c) rain on Mon & Tues w/ daily $P(\text{rain}) = 0.3$
 $P(M \& T) = (0.3)(0.3)$
 d) 5 blue & red, draw blue then red no replace
 $P(B \& R) = P(B)P(B|R) = \left(\frac{5}{10}\right)\left(\frac{4}{9}\right)$

percents

$\% \Delta = \frac{\text{difference}}{\text{original}} \times 100$
 $\% \text{ less than} \rightarrow \text{Final} = 1 - \frac{\% \text{ less than}}{100} \times \text{initial}$
 $\% \text{ greater than} \rightarrow \text{Final} = 1 + \frac{\% \text{ less than}}{100} \times \text{initial}$
 $\% \text{ discount} = \text{price}(1 - \text{percentage})$
 $\% \text{ mark-up} = \text{price}(1 + \text{percentage})$

examples:
 a) 12 is 5% of what $\rightarrow 0.05x = 12$
 b) 7% of 9 is what percent of 7 $\rightarrow 0.07(9) = \frac{x}{100}(7)$
 c) 200% of 4% of 50 is what $\rightarrow \frac{200}{100} \times \frac{4}{100} \times 50$
 d) what percent of y% of 50 is 40% of y $\rightarrow \frac{x}{100} \times \frac{y}{100} \times 50 = \frac{40}{100}y$
 y cancel out

interest

simple interest $= (\text{principal})(\text{rate})(\text{time})$
 compound interest: $a = p(1 + \frac{r}{n})^{nt}$
 $a \rightarrow$ future value, $p \rightarrow$ principal, $r \rightarrow$ rate, $t \rightarrow$ time
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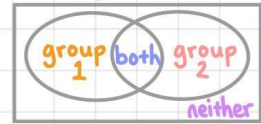
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sequences

arithmetic sequence: $a_n = a_1 + (n-1)d$

- ↳ n^{th} term
- ↳ first term
- ↳ term #
- ↳ common difference
- ↳ # of terms added

$$\Sigma \rightarrow S_n = \frac{n}{2}(a_1 + a_n) \leftarrow \text{sum of first } n \text{ terms}$$

geometric sequence: $a_n = a_1 r^{(n-1)}$

- ↳ n^{th} term
- ↳ first term
- ↳ term #
- ↳ common ratio

work + rates algebra

distance/work = (rate)(time)

work = (individual rate)(# individuals)(time)

average speed = total distance ÷ total time

for chase problems → subtract rates

proportion → equivalent relationships b/w ratio
keep track units

exponents:

$$x^0 = 1$$

$$x^a x^b = x^{a+b}$$

$$x^{-a} = \frac{1}{x^a}$$

$$\frac{x^a}{x^b} = x^{a-b} = \frac{1}{x^{b-a}}$$

$$x^a y^a = (xy)^a$$

$$\left(\frac{x}{y}\right)^a = \frac{x^a}{y^a}$$

$$(x^a)^b = x^{ab}$$

$$\text{PEMDAS}$$

roots:

$$\sqrt[n]{x^m} = x^{m/n}$$

$$\sqrt[n]{x} \sqrt[n]{y} = \sqrt[n]{xy}$$

$$\sqrt[n]{\frac{x}{y}} = \frac{\sqrt[n]{x}}{\sqrt[n]{y}}$$

$$\sqrt[n]{x^m} = x^{m/n}$$

$$\sqrt[n]{x} = x^{1/n}$$

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difference of squares:

$$x^2 - y^2 = (x+y)(x-y)$$

$$(x+y)^2 = x^2 + 2xy + y^2$$

$$(x-y)^2 = x^2 - 2xy + y^2$$

$$\text{expanded}$$

$$\text{absolute value:}$$

$$|x+y| \leq |x| + |y|$$

$$|x-y| \geq |x| - |y|$$

$$|x+y| = z \rightarrow \pm(x+y) = z$$

$$|x-y| = z \rightarrow \pm(x-y) = z$$

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divisibility:

a) # divide by 3 → Σ digits ÷ 3

b) # divide by 4 → last 2 digits ÷ 4

c) # divide by 6 → also ÷ by 2 + 3

d) # divide by 8 → last 3 digits ÷ 8

e) # divide by 9 → Σ digits ÷ 9

$$\frac{x}{y} = \text{quotient} + \frac{\text{remainder}}{y}$$

ratio

ratios → part to whole relationship

	parts	total
ratio		ratio total
multiplier		
actual		actual total

whole → ratio total = Σ all parts

$$\text{multiplier} = \frac{\text{actual total}}{\text{ratio total}}$$

$$\text{partial actual} = (\text{part})(\text{multiplier})$$

each part has column

examples:

a) flour, egg, yeast, salt mixed in ratio of 11:9:3:2, how many lbs of yeast in 20 pound mix

	flour	egg	yeast	salt	total
ratio	11	9	3	2	25
multiplier	4/5	4/5	4/5	4/5	4/5
actual			12/5		20

$$\text{multiplier} \rightarrow 25 \times \frac{4}{5} = 20$$

$$\text{yeast} \rightarrow 3 \times \frac{4}{5} = \frac{12}{5}$$

composite #s → non-prime

2, 3, 5, 7, 11, 13...

prime factorization: breaking down # into product of prime #s → integers w/ factors 1 + itself

prime factorize each, use exponents

LCM for repeated prime factors take ones w/ largest exponent, w/ same power just take once

multiply previous w/ all non-repeated prime factors

prime factorize each, use exponents

GCF for repeated prime factors take ones w/ smallest exponent, w/ same power take once

multiply previous

factors prime factorize, use exponents

+1 to each exponent + multiply each

no repeated means GCF = 1

examples:

a) LCM 24 + 60

$$\text{LCM} \rightarrow 2^3 \times 3^1 \times 5^1 = 120$$

b) GCF 24 + 60

$$2^2 \times 3^1 = 12$$

c) # factors of 240

$$(4+1)(1+1)(1+1) = 20$$

tree

$$\begin{array}{c} 4 \quad 6 \quad 10 \\ \swarrow \quad \downarrow \quad \searrow \\ 2 \quad 2 \quad 2 \quad 3 \quad 5 \\ \swarrow \quad \downarrow \quad \searrow \\ 2^3 \times 3^1 \quad 2^2 \times 3^1 \times 5^1 \end{array}$$

prime factorization tree

$$2^2 \times 3^1 = 12$$

$$(4+1)(1+1)(1+1) = 20$$

$$2^4 \times 3^1 \times 5^1$$

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prime factorization

statistics

$$\text{average} = \frac{\text{total } \Sigma}{\text{\# of things}}$$

weighted average:

$$\frac{(x_1)(f_1) + (x_2)(f_2) + \dots}{\Sigma f}$$

$$\Sigma f \rightarrow \text{total freq of data points}$$

standard deviation:

a) when add/subtract all terms by constant → s.d. no Δ

b) when multiply/divide all terms by constant → s.d. Δ by same amount

measure of spread

evenly spaced set → mean = median

median: middle value, numerical order

VOCABULARY

READING

a-words

adjectives
adverbs
verbs
nouns

Abate: to lessen in intensity
Aberrant: abnormal
Abreast: informed
Abridge: to shorten
Abscond: to depart secretly
Accolade: award, honor
Acerbic: bitter
Acrimonious: angry
Acumen: accurate insight
Adept: skilled
Admonish: to advise
Adulation: excessive praise
Adulterate: to spoil
Advocate: to support
Aesthetic: pertaining to beauty
Agglomerate: to lump
Aggrandize: to increase influence
Aghast: shocked
Agog: eagerness
Alacrity: eagerness
Amalgamate: to combine
Ambivalence: hesitant
Ameliorate: to make better
Amenable: agreeable
Anachronism: something not belonging to time period
Anomaly: abnormality
Antedate: to predate
Anodyne: inoffensive
Antipathy: hatred
Apace: quickly
Apathy: lack of feeling
Approbation: expression of approval
Archaic: ancient
Arduous: strenuous
Artless: genuine
Ascertain: to investigate
Ascetic: self discipline
Asperity: harshness
Assuage: to pacify
Astringent: sharp, severe
Astringent: substance making skin tight
Audacious: daring
Auspicious: lucky
Austere: simple
Avarice: greed
Aver: to declare
Axiomatic: universally accepted

b-words

Banal: boring
Belligerent: eager to fight
Bolster: to support
Bombastic: arrogant
Bucolic: rural
Burgeon: to flourish
Cacophony: harsh sound
Canon: established set of principles
Canonical: agrees with principles
Capricious: changing mind quickly and often
Castigation: severe punishment
Catalyst: thing that speeds up result
Caustic: corrosive
Censure: to criticize
Chary: cautious, wary
Chicanery: deceitfulness
Coalesce: to combine
Cogent: convincing
Complaisance: willing to comply
Connoisseur: judge, critic
Contentious: causing disagreement
Contiguous: side-by-side
Contrite: regretful
Convention: agreement
Convolutéd: complicated
Corroborate: to confirm
Credulous: gullible
Culpable: blamable
Cynicism: belief that people are motivated by selfishness

adjectives
adverbs
verbs
nouns

d-words

Dearth: scarcity
Decorum: proper behavior
Demur: to question
Deride: to ridicule
Derision: mockery
Desiccate: to dry
Diatribé: denunciation
Didactic: instructive
Diffident: timid
Dilettante: amateur
Disabuse: to undeceive
Discordant: harsh sound
Discretion: cautious reserve in speech
Disinterested: free of bias
Disparage: to belittle
Disparate: distinct
Dissemble: to mislead
Dissonance: lacking harmony
Divulge: to reveal
Dogmatic: stubbornly opinionated

e-words

Ebullience: cheerfulness
Eccentric: unconventional
Eclectic: choosing from wide range
Effrontery: boldness
Elegy: sad poem
Eloquent: well-spoken
Emollient: soothing
Empirical: based on observation or experiment
Endemic: native to location
Enervate: to weaken
Engender: to cause
Enigma: puzzle
Enigmatic: mysterious
Enmity: hostility
Ennui: dissatisfaction from boredom
Ephemeral: short-lived
Equivocal: ambiguous
Equivocate: to use ambiguous language for deception
Erudite: scholarly
Esoteric: restricted to small group
Eulogy: speech honoring dead
Euphony: pleasant sound
Evanescent: vanish like vapor
Exacerbate: to make worse
Excoriate: to criticize
Exculpate: to prove innocent
Exigent: urgent
Exonerate: to prove innocent
Extant: surviving
Extemporaneous: improvised

adjectives
adverbs
verbs
nouns

f-words

Facetious: inappropriate humor
Fallacy: mistaken belief
Fawn: to flatter
Fervent: emotional
Fervid: really passionate
Filibuster: long speech to delay legislature
Flout: to mock
Fortuitous: fortunate
Fulminate: to protest
Furtive: secretive

g-words

Garrulous: talkative
Germane: relevant
Gibe: insulting taunt
Glib: insincere speech, superficial
Gossamer: delicate
Grandiloquence: arrogant speech
Gregarious: sociable
Gyrate: to spiral

h-words

Hackneyed: cliché
Halcyon: calm
Harangue: long ranting speech
Harangue: to deliver a ranting speech
Hedonism: pursuit pleasure
Headlong: with reckless haste
Hector: to harass
Hegemony: domination over others
Heretical: violation
Homogenous: similar
Hubris: prideful

adjectives
adverbs
verbs
nouns

i-words

Iconoclast: rebel
Idolatrous: excessive devotion
Imminent: about to happen
Immutable: unchangeable
Impassive: unemotional
Impecunious: poor
Imperturbable: emotionally steady
Impetuous: impulsive
Implacable: unable to be calmed
Impunity: freedom from punishment
Inchoate: undeveloped
Incipient: begin to exist
Inculcate: to accuse
Indifferent: having no preference
Inert: inactive
Infelicitous: inappropriate
Ingenuous: unsuspecting
Inimical: hostile
Innocuous: harmless
Insidious: lacking interest
Intractable: stubborn
Intransigent: refusing to compromise
Intrepid: brave
Inured: accustomed to accepting something undesirable
Inveigle: to obtain by deception
Irascible: easily angered

l-words

Lachrymose: sad
Laconic: using few words
Largess: generous
Laud: to praise highly
Laudable: deserving praise
Libidinous: lustful
Loquacious: talkative
Lucid: clear
Ludic: playful in an aimless way
Lugubrious: sad
Luminous: bright

m-words

Macabre: disturbing
Magnanimity: forgiving
Malevolent: spiteful
Malleable: capable of being shaped
Martial: warlike
Martinet: strict disciplinarian
Maudlin: very emotional
Maverick: nonconformist, independent
Mendacity: dishonesty
Mercenary: greedy
Mercurial: sudden unpredictable change
Meticulous: detailed and thorough
Minatory: threatening
Misanthrope: hater of mankind
Mitigate: to lessen
Mollify: to soothe
Mordant: hard-hearted
Morose: sad
Mundane: ordinary, worldly
Munificent: very generous

adjectives
adverbs
verbs
nouns

n-words

Nadir: all time low
Nascent: beginning to exist, develop
Nebulous: cloudy, uncertain
Neologism: new word or expression
Neophyte: beginner
Noisome: unpleasant
Nonplussed: perplexed
Novitiate: beginner in religious order
Noxious: harmful

o-words

Obdurate: stubborn
Obfuscate: to make confusing
Objurgate: to scold
Obloquy: strong criticism, verbal abuse
Obsequious: eager to please
Obstinate: stubborn
Obtuse: mentally dull
Obviate: to anticipate and prevent
Occlude: to block
Odious: repulsive
Onerous: burdensome
Opaque: unable see through, hard to understand
Opprobrium: harsh criticism
Optimum: most favorable outcome
Orifice: opening
Oscillation: movement back-and-forth, hesitation
Ostensible: stated to be true, but not necessarily so
Ostentation: excessive showiness
Ostentatious: showy

p-words

adjectives
adverbs
verbs
nouns

Paeon: song of praise
Paucity: scarcity
Pedagogy: art of teaching
Pedant: someone who overly shows off learning
Pellucid: clear
Penurious: stingy
Penury: poverty
Perennial: lasting long time
Perfidy: treachery
Perfunctory: done without care
Pernicious: gradual harmfulness
Perspicacious: wise
Peruse: to examine carefully
Pervade: to spread throughout
Pervasive: spread throughout
Philistine: person lack culture
Phlegmatic: unemotional, sluggish
Pine: to yearn for
Pious: religious
Pirate: to illegally use or reproduce
Pith: essential part
Pithy: precise
Placate: to soothe
Platitude: cliché
Plethora: surplus
Polemic: verbal attack
Polemical: controversial, argumentative
Pragmatic: practical rather than idealistic
Prattle: to babble meaninglessly
Precipitate: acting hastily with impulse
Precipitate: to cause to happen suddenly
Predilection: preference
Preen: to dress up
Prescience: clairvoyant
Presumptuous: disrespectfully bold
Prevaricate: to speak/act in an evasive way, beat-around-the-bush
Pristine: pure, uncorrupted
Probity: honesty
Proclivity: natural predisposition
Prodigal: wastefully lavish
Prodigious: enormous
Profligate: wastefully lavish
Profuse: abundant
Proliferate: to increase quickly
Prolific: productive
Propensity: natural predisposition
Propriety: being proper
Prosaic: dull
Puerile: childish
Putrefy: to rot

q-words

Quaff: to drink deeply
Qualms: causes for hesitancy
Queries: questions
Querulous: complaining
Quiescence: stillness
Quixotic: idealistic but impractical
Quotidian: recurring daily

adjectives
adverbs
verbs
nouns

r-words

Raconteur: skilled storyteller
Rancorous: resentment
Rarefy: to make thinner
Recalcitrant: disobedient
Recant: to retract from previous beliefs
Recondite: difficult to understand
Redoubtable: awe-inspiring
Refulgent: shiny
Relegate: to assign to lower position
Reneg: to go back on a promise
Repudiate: to reject
Rescind: to cancel
Restive: restless
Reticent: silent/reserved
Reverent: deeply respectful
Rhetoric: effective writing/speaking

s-words

Salubrious: promoting health
Sanction: official permission
Sanction: to authorize
Satire: literary work ridiculing human misconduct and stupidities
Sedulous: hard working
Shard: broken fragment
Solicitous: concerned, eager
Solvent: able to dissolve other substances, able to pay debts
Soporific: sleep-inducing
Sordid: filthy
Specious: seemingly plausible, but actually wrong
Spendthrift: who wastes money
Sporadic: occasionally
Spurious: not genuine
Squalid: filthy
Squander: to spend wastefully
Stoic: indifferent to pleasure/pain
Stupefy: to stun
Stymie: to hinder
Subpoena: court order appearance
Succinct: concise
Superfluous: extra, unnecessary
Supplant: to replace
Surfeit: extra
Synthesis: combination

t-words

Tacit: implied, not explicit
Temperate: showing moderation/self-restraint
Tenacity: persistence
Tenuous: flimsy
Terse: concise
Timorous: fear, lack of confidence
Tirade: angry speech
Torpid: sluggish
Torque: rotational force
Tortuous: complicated
Transient: temporary
Trenchant: insightful
Trepidation: apprehension
Trite: cliché
Trout: to praise
Truculent: eager to fight
Truism: true statement, but not interesting or new
Turgid: swollen, tediously pompous
Turbid: obscure, cloudy
Tyro: beginner

adjectives
adverbs
verbs
nouns

u-words

Ubiquitous: widespread
Unfeigned: genuine
Untenable: indefensible
Urbane: courteous, civilized

v-words

Vacillate: to sway indecisively
Variegated: multicolored
Venal: greedy
Venerate: to look up to
Veracity: honesty
Verity: truth
Vexation: annoyance
Vicarious: indirect empathy
Vigilant: watchful
Vilify: to defame
Vindictive: desire revenge
Virulent: bitterly hostile
Viscous: thick, sticky
Vitiate: to weaken
Vitriolic: malice
Vituperate: to condemn
Volatile: unstable
Voracious: huge appetite

w-words

Waver: to fluctuate between choices

z-words

Zeal: strong feeling of interest
Zealous: enthusiastically devoted